Vital Balance: Homeostasis and Allostasis

Homeostasis

Maintenance of stability.

Regulation of the internal environment (temperature, pH, blood glucose).

Allostasis

Capacity for adaptation.

Maintenance of homeostasis under changing conditions.

Stress

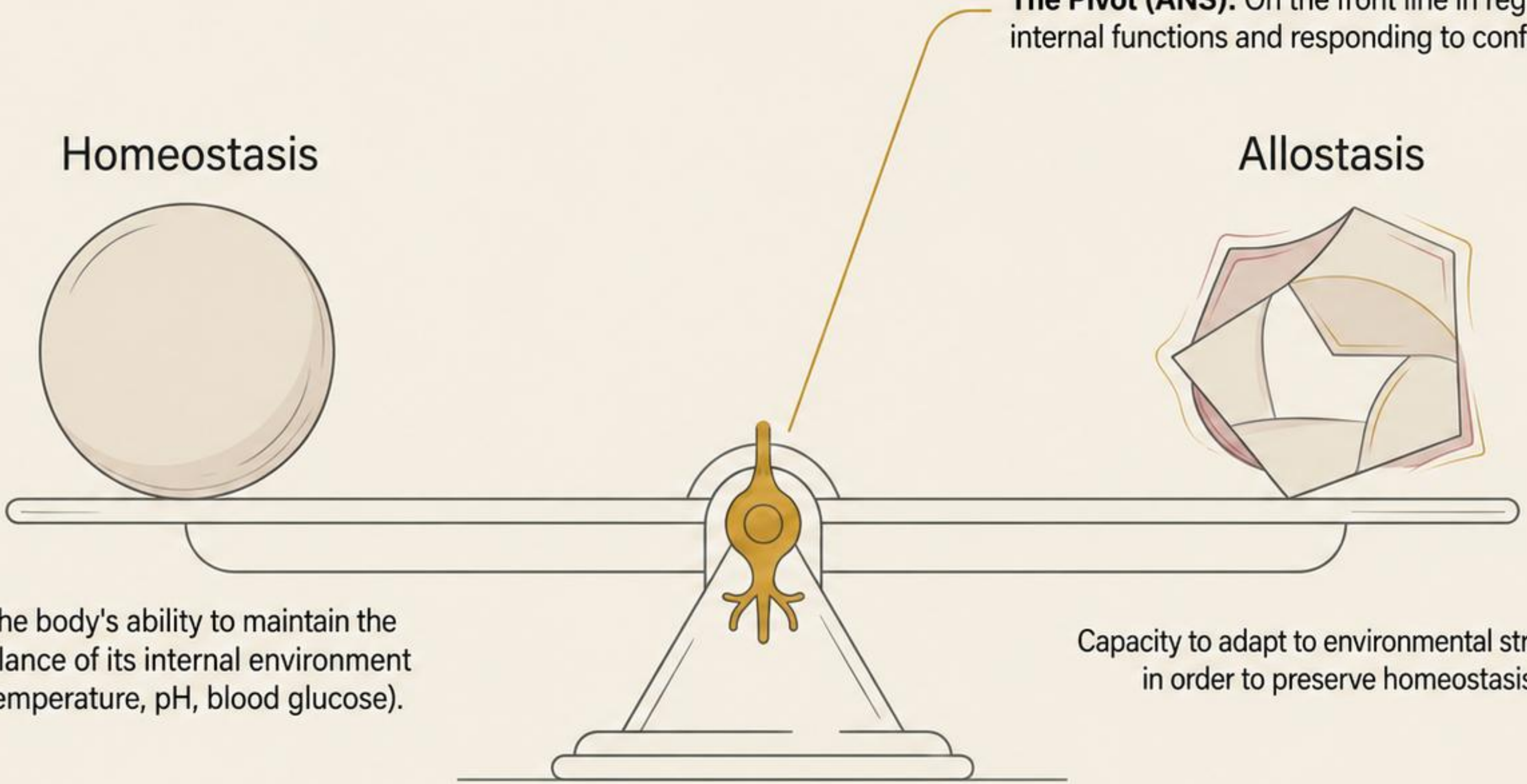
Protective response.

Set of responses to a threat aimed at limiting homeostatic disruption.

Key role of the ANS: The Autonomic Nervous System plays a front-line role in preserving internal balance.

The Stress Mechanism: Stability in Change

The Pivot (ANS): On the front line in regulating internal functions and responding to conflict situations.



Homeostasis

The body's ability to maintain the balance of its internal environment (temperature, pH, blood glucose).

Allostasis

Capacity to adapt to environmental stressors in order to preserve homeostasis.

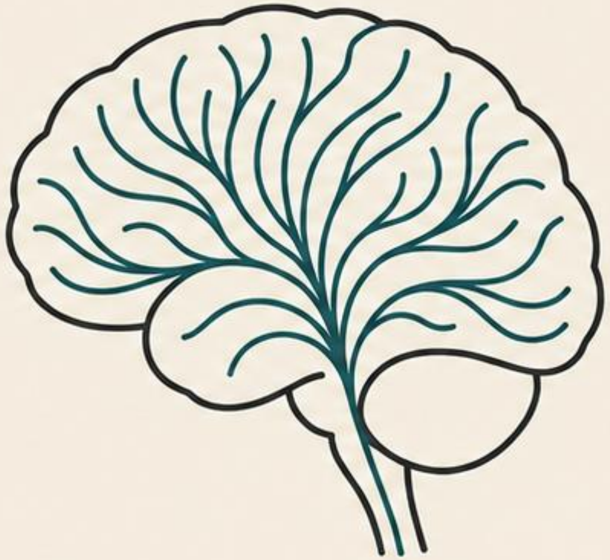
Stress is a fundamental protective phenomenon. In the short term, it protects. In the long term, it carries an energy cost.

Duality of Stress: Distress vs. Eustress

Distress (Negative)

- **Emotions:** Fear, anger, insecurity, failure.
- **Neuroendocrine response:** Cortisol dominance.
- **Impact:** Weakening of the nervous, hormonal and immune systems. Pathogenic progression.

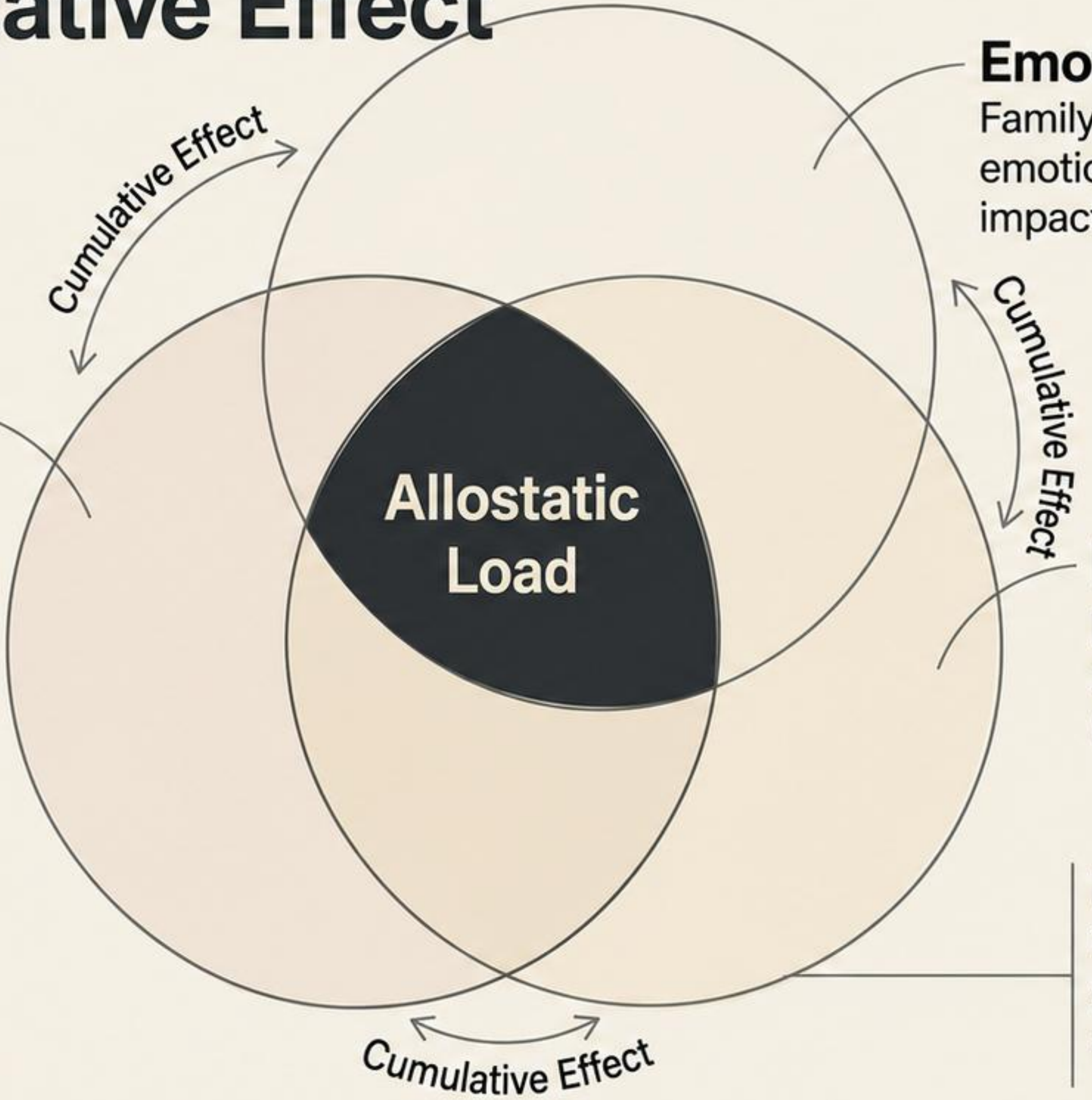
Eustress (Positive)



- **Emotions:** Joy, love, enthusiasm, security.
- **Neuroendocrine response:** Neuromediator activation (adrenaline).
- **Impact:** Favorable adaptive response, stimulating and necessary in the short term.

Differentiation of the polarity of stress and its hormonal response.

The Trilogy of Stressors and the Cumulative Effect



Emotional:
Family conflicts, isolation, emotional shocks (major impact on health).

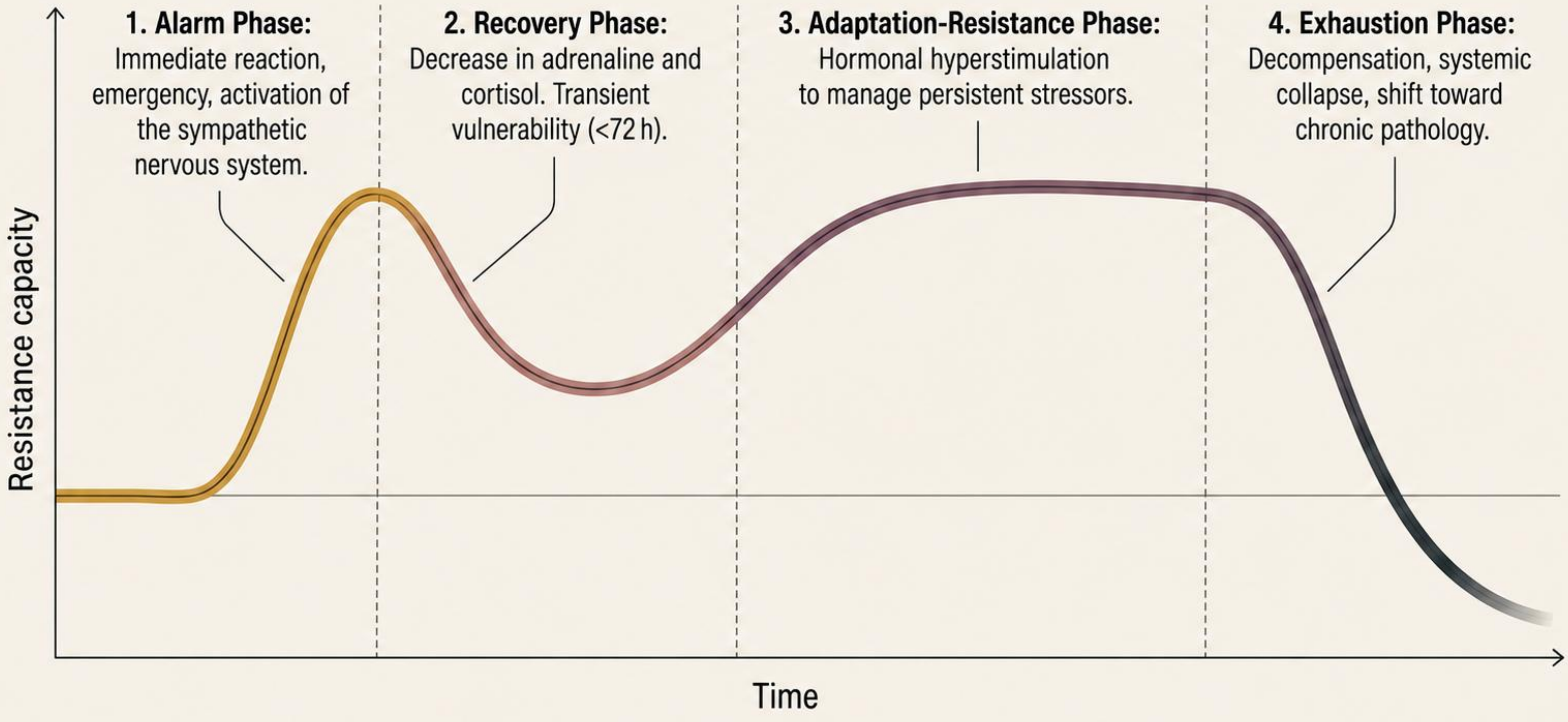
Physical:
Accidents, sleep deprivation, chronic pain, sensory stressors.

Biochemical:
Dysbiosis, food intolerances, heavy metals (often act insidiously).

Emotional stress can influence the gastrointestinal tract, generating spinal pain and ultimately affecting overall biology.

The General Adaptation Syndrome (GAS)

Model developed
by Hans Selye.

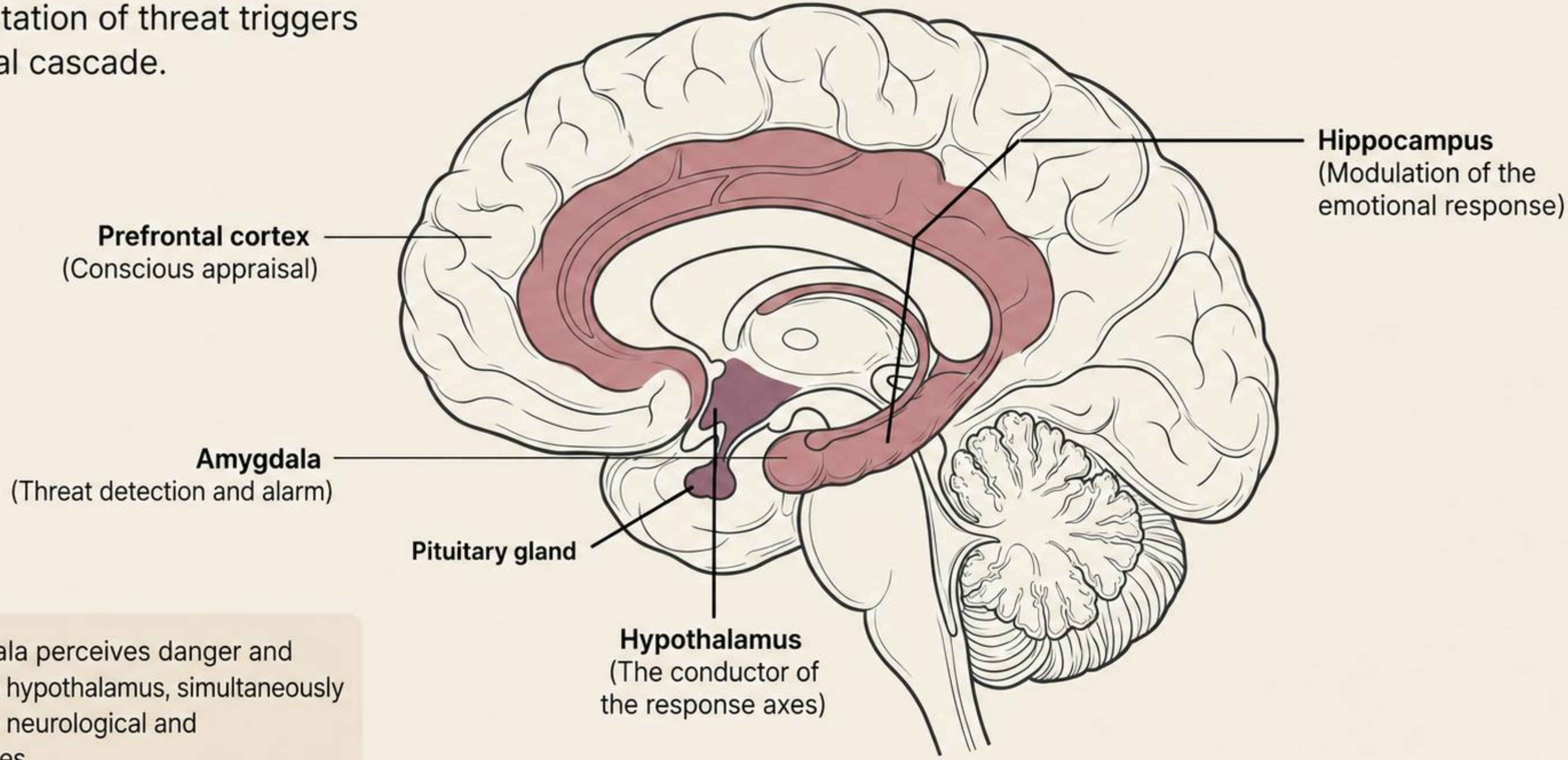


Alarm phase: Physiological response matrix

SAM Axis (Neurological)	HPA Axis (Hormonal)
- Speed: Immediate (Emergency) - Origin: Hypothalamus → Locus coeruleus (Brainstem) - Key hormones: Adrenaline / Noradrenaline - Physiological effects: Increased heart rate, increased blood pressure, blood flow to striated muscles (fight or flight). Digestion put on standby.	- Speed: Secondary (if stress persists) - Origin: Hypothalamus (CRH) → Anterior pituitary (ACTH) - Key hormones: Cortisol, Aldosterone, DHEA - Physiological effects: Increased availability of glucose/cholesterol, renal reabsorption, immune modulation.

The Tipping Point: Limbic Brain and Hypothalamus

The interpretation of threat triggers the biological cascade.



The amygdala perceives danger and activates the hypothalamus, simultaneously initiating the neurological and hormonal axes.

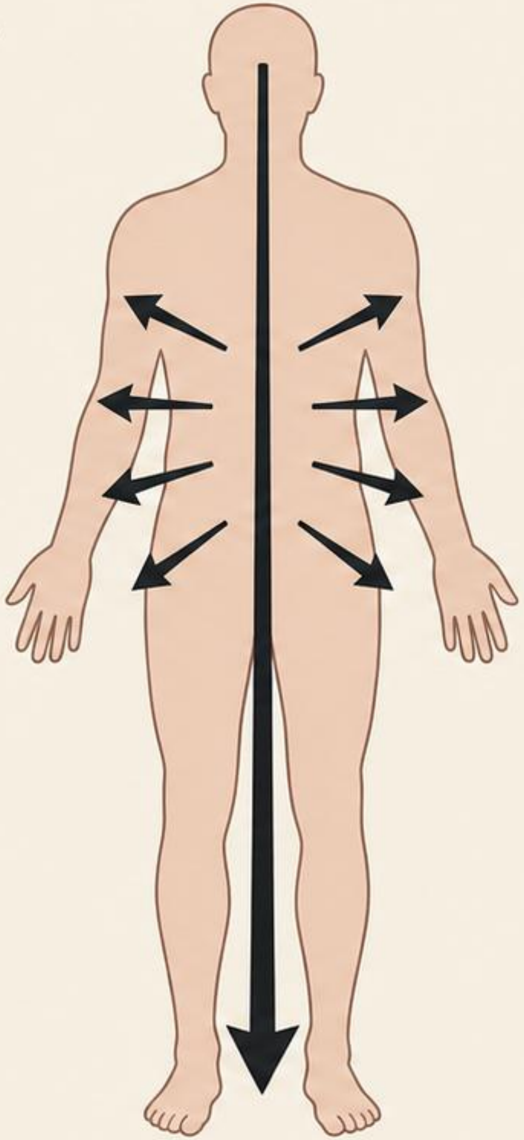
Phase 2: Recovery and Vulnerability

Keywords: Fatigue, recovery, transient vulnerability (max 72 h).

Physiology: Decrease in cortisol and adrenaline.
Recrudescence of pain, increased emunctory elimination.

Therapeutic Relevance (Hering's Law) :
In reflexotherapy, this phase explains post-session reactions. Healing proceeds:

1. From top to bottom
2. From within outward
3. From chronic to acute



Period of physiological recovery and direction of salutary symptoms.

Phase 3: Adaptation-Resistance

Keywords: Organization of defenses, chronicity, persistent stressors.

Hyperstimulation (Support)

Adrenal glands: Excess cortisol (hypertension, osteopoross).

Pancreas: Hyperinsulinemia (cellular resistance).

Thyroid: Excess T3/T4 (nervousness, tachycardia).

Hypostimulation (The shift)

Progression toward hypothyroidism, hypoglycemia, and hypocortisolism (orthostatic dizziness).

Pathological Status: Onset of allodynia (exacerbated pain). Anatomical integrity begins to give way.

The metabolic cost of prolonged adaptation.

Exhaustion: From Dysfunction to Chronicity

Loss of anatomical integrity

	Functional (Reversible)	Chronic (Irreversible)
Immune System	Mild allergies, food intolerances.	Severe infectious diseases, autoimmune diseases.
Digestive System	Hypersensitivity, bowel transit disorders, dysbiosis, irritable bowel syndrome.	Severe gastroduodenal ulceration.
Nervous / Limbic System	Fear, anxiety, allodynia (abnormal joint pain).	Major depression, hippocampal atrophy, Alzheimer's disease, neurodegenerative diseases.

« We are ill because we lose our health, not the other way around. »

The failure of adaptive mechanisms in the face of allostatic load is the antechamber to pathology.

Clinical Conclusion: Functional and integrative therapies (ROP) do not have the sole aim of suppressing a symptom, but rather of supporting the emunctory organs and restoring the vitality necessary for the body to fight disease.